CSE111: Programming Language II (Lab)
Spring 2026
Lab Exam - 01
Duration:30 Minutes

A

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

You are planning to serve some customers in a particular queue. You will assign priority score to each of the customers based on the sum of ascii values for characters in their respective names. You will always serve the customer with highest priority first.

<u>For example:</u> Sam's priority score will be sum of ascii code for 'S', 'a' and 'm' which is 83 + 97 + 109 = 289

Now Design the **Service** class to generate the desired output shown below.

Tester Code	Output

`public class MaintainQueue{`
`  public static void main(String[] args) {`
 	
`  Service queue = new Service();`
`  queue.serve();`
 	
`  System.out.println("1==========");`
 	
`  queue.initiate = true;`
`  queue.customers = new String[]{"Sam","Jes", "Alo", "Era"};`
`  queue.priorityScore();`
 	
`  System.out.println("2==========");`
 	
`  queue.checkQueue();`
 	
`  System.out.println("3==========");`
 	
`  queue.serve();`
`  queue.checkQueue();`
 	
`  }`
`}`
	No customer in the queue
	1==========
	priority scores assigned
	2==========
	Customers in queue:
	Sam - priority score 289
	Jes - priority score 290
	Alo - priority score 284
	Era - priority score 280
	3==========
	Currently serving Jes
	Customers in queue:
	Sam - priority score 289
	Alo - priority score 284
	Era - priority score 280

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B

Name:	ID:	Section:

Question 1 [CO1] [10 Points]

You are planning to interview a batch of candidates. You will assign an interview ID to each of the candidates based on the sum of ascii values for characters in their respective names. You will always interview the candidate that has the largest value of ID first.

<u>For example:</u> Zoe's interview ID will be sum of ascii code for 'Z', 'o' and 'e' which is 90 + 111 + 101 = 302

Now Design the **Interview** class to generate the desired output shown below.

Tester Code	Output

`public class ConductInterview{`
` public static void main(String[] args) {`
 	
`  Interview batch = new Interview();`
`  batch.call();`
 	
`  System.out.println("1==========");`
 	
`  batch.start = true;`
`  batch.candidates = new String[]{"Zoe","Eva", "Kia","Ivy"};`
`  batch.assignSerial();`
 	
`  System.out.println("2==========");`
 	
`  batch.checkCandidates();`
 	
`  System.out.println("3==========");`
 	
`  batch.call();`
`  batch.checkCandidates();`
 	
` }`
`}`
	No candidate to interview
	1==========
	Interview ID assigned
	2==========
	Upcoming Candidates:
	Zoe - ID number 302
	Eva - ID number 284
	Kia - ID number 277
	Ivy - ID number 312
	3==========
	Currently interviewing Ivy
	Upcoming Candidates:
	Zoe - ID number 302
	Eva - ID number 284
	Kia - ID number 277

Solutions

Set A solution:

```
public class Service{
    String [] customers;
    int [] score = new int [4];
    boolean initiate;

    public void priorityScore(){
        for(int i=0; i<customers.length; i++){
            for(int j=0; j<customers[i].length(); j++){
                score[i] += (int)customers[i].charAt(j);
            }
        }
        System.out.println("priority scores assigned");
    }
    public void serve(){
        if(initiate){
        int max = 0;
        int max_index = 0;
        for(int i=0; i<score.length; i++){
            if(score[i]>max){
                max = score[i];
                max_index = i;
            }
        }
        System.out.println("Now serving "+customers[max_index]);
        customers[max_index] = null;
        score[max_index] = 0;
        }
        else{
            System.out.println("No customer in the queue");
        }
    }
    public void checkQueue(){
        System.out.println("Customers in queue:");
        for(int i=0; i<score.length; i++){
            if(customers[i]!=null){
                System.out.println(customers[i]+": priority score "+score[i]);
            }
        }
    }
}
```

Set B solution:

```
public class Interview{
    String [] candidates;
    int [] score = new int [4];
    boolean start;

    public void assignSerial(){
        for(int i=0; i<candidates.length; i++){
            for(int j=0; j<candidates[i].length(); j++){
                score[i] += (int)candidates[i].charAt(j);
            }
        }
        System.out.println("Interview ID assigned");
    }
    public void call(){
        if(start){
        int max = 0;
        int max_index = 0;
        for(int i=0; i<score.length; i++){
            if(score[i]>max){
                max = score[i];
                max_index = i;
            }
        }
        System.out.println("Currently interviewing "+candidates[max_index]);
        candidates[max_index] = null;
        score[max_index] = 0;
        }
        else{
            System.out.println("No candidate to interview");
        }
    }
    public void checkCandidates(){
        System.out.println("Upcoming Candidates:");
        for(int i=0; i<score.length; i++){
            if(candidates[i]!=null){
                System.out.println(candidates[i]+" - ID number "+score[i]);
            }
        }
    }
}
```

Marking Rubric

For both sets:

Criteria	Points
Basic Class Design → [1] Correct class declaration → [0.5 + 0.5] 2 instance arrays (candidates/customers, score) and boolean variable named “start”	2
Method: **assignSerial()**:void / **priorityScore()**:void → [1.5] declaring the method → [0.5] outer loop to trace the names → [0.5] inner loop to trace the characters in a name → [0.5] calculate sum of ASCII code → [0.5] assign the sum to associated index	3.5
Method: **call()**:void / **serve()**:void → [1] declaring method → [1] if-else to check if we started interviewing / serving → [0.5] loop to calculate + store maximum priorityScore / IDNumber → [0.5] delete the name and score of when interview call / customer serving is done.	3
Method: **checkCandidates()**:void / **checkQueue()**:void → [0.5] declaring method → [0.5] Initial printing statement → [0.5] print the non-null candidates	1.5
Total:	**10**